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Yamada

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(54) **INFORMATION PROCESSING TERMINAL,
INFORMATION PROCESSING METHOD,
AND PROGRAM**

(58) **Field of Classification Search**

CPC H04N 21/47217; H04N 21/4325; H04N
21/439; H04N 21/44

See application file for complete search history.

(71) Applicant: **SATURN LICENSING LLC**, New
York, NY (US)

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(72) Inventor: **Sohei Yamada**, Kanagawa (JP)

(73) Assignee: **SATURN LICENSING LLC**, New
York, NY (US)

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(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

This patent is subject to a terminal dis-
claimer.

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(21) Appl. No.: **18/238,333**

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010744 mailed May 26, 2020. pp. 1-3.

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Primary Examiner — Mulugeta Mengesha

(74) *Attorney, Agent, or Firm* — Richard LaPeruta

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(63) Continuation of application No. 17/440,457, filed as
application No. PCT/JP2020/010744 on Mar. 12,
2020, now Pat. No. 11,778,275.

(57) **ABSTRACT**

The present technology relates to an information processing
terminal, an information processing method, and a program,
each of which makes it possible to easily switch a setting
regarding playback of content.

(30) **Foreign Application Priority Data**

Mar. 26, 2019 (JP) 2019-059130

An information processing terminal according to an aspect
of the present technology starts playback of content by using
a default value as a setting value of a predetermined setting
regarding playback, and controls the playback of the content
so that, in a case where a switching operation for switching
the predetermined setting is performed, the content is played
back by using a setting value selected during previous
playback. The present technology is applicable to a smart-
phone in which various applications can be installed to
extend a function thereof.

(51) **Int. Cl.**

H04N 21/472 (2011.01)

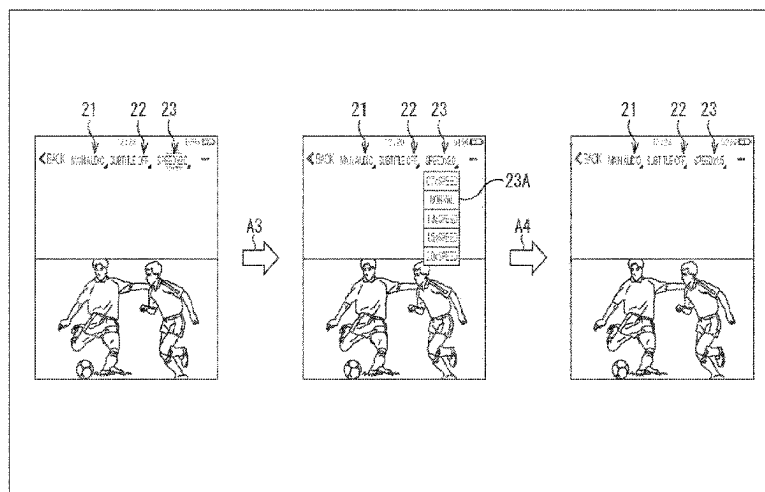
H04N 21/432 (2011.01)

(Continued)

(52) **U.S. Cl.**

CPC ... **H04N 21/47217** (2013.01); **H04N 21/4325**
(2013.01); **H04N 21/439** (2013.01); **H04N**
21/44 (2013.01)

20 Claims, 12 Drawing Sheets



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H04N 21/439 (2011.01)
H04N 21/44 (2011.01)

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FIG. 1

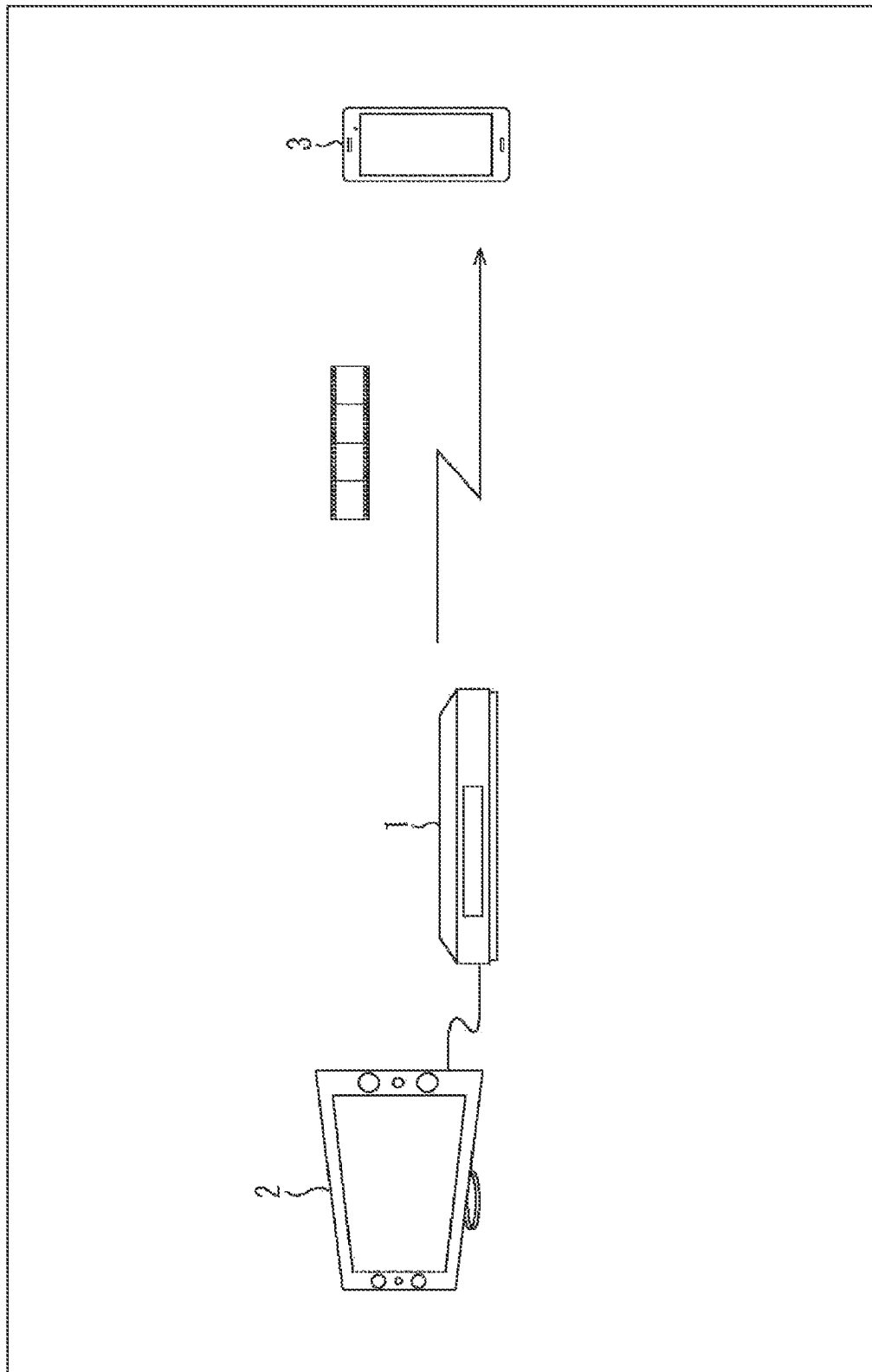


FIG. 2

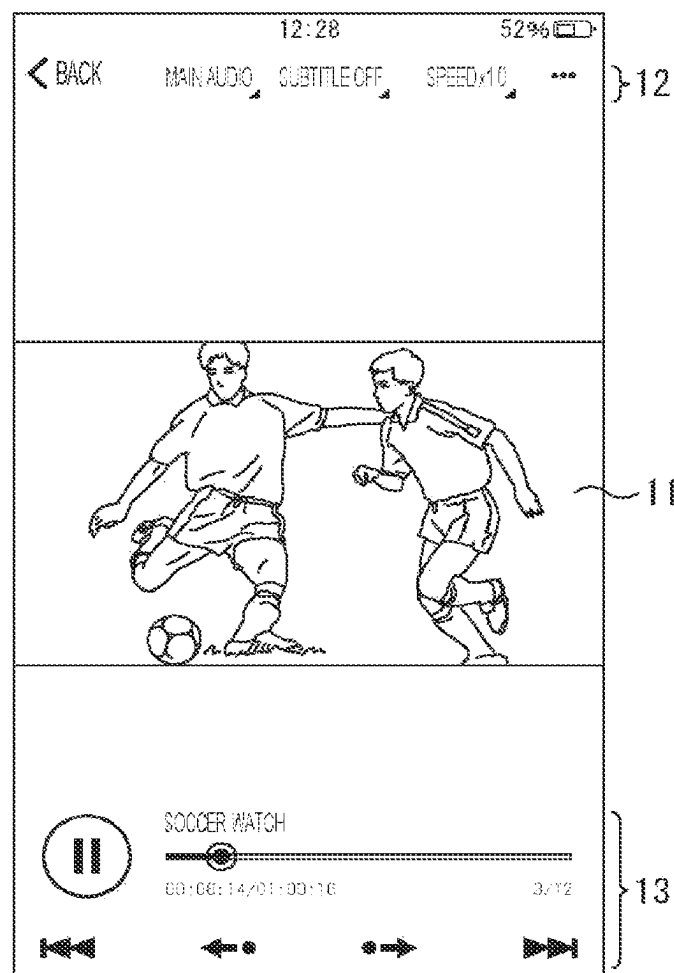


FIG. 3

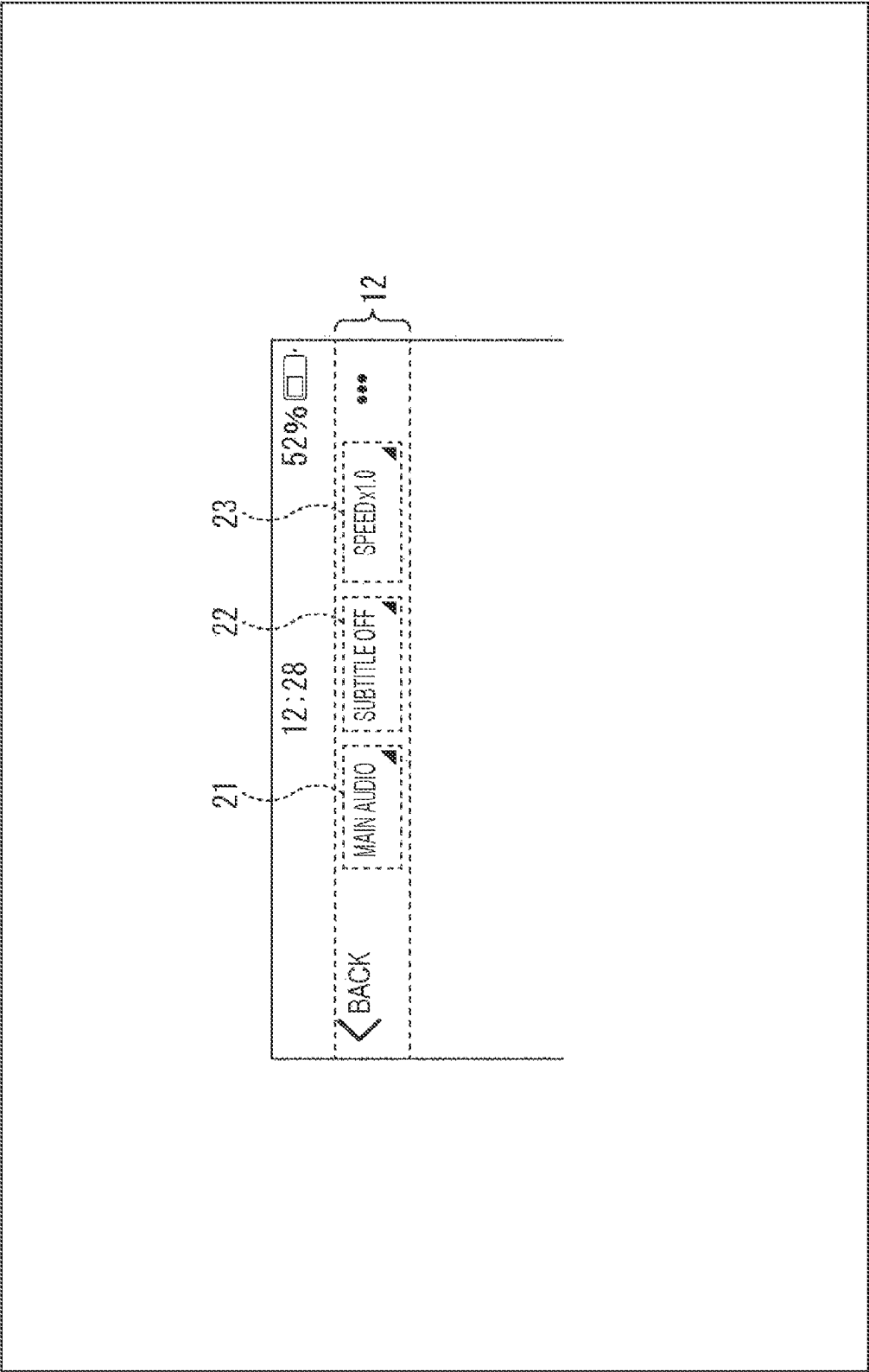


FIG. 4

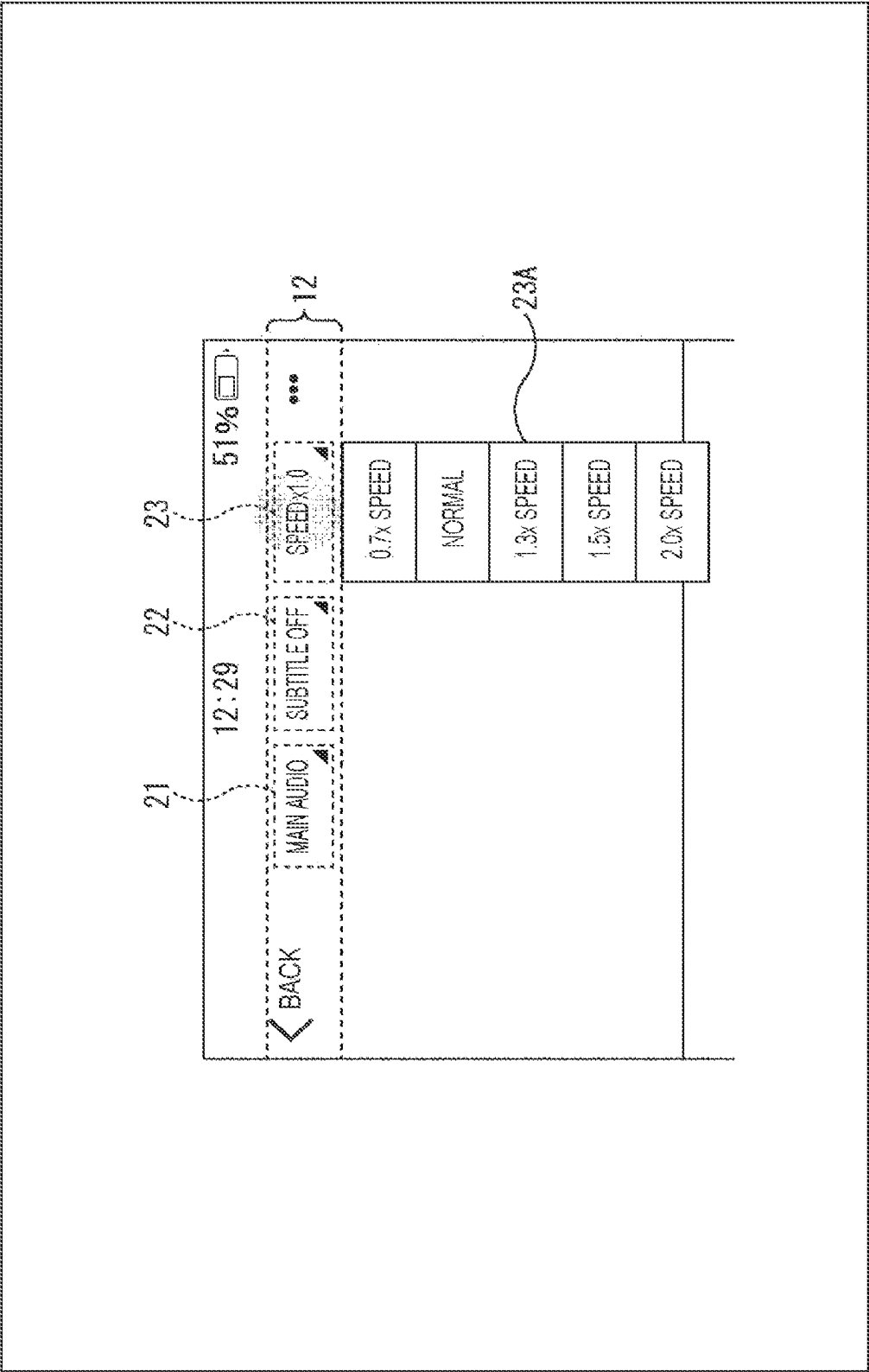


FIG. 5

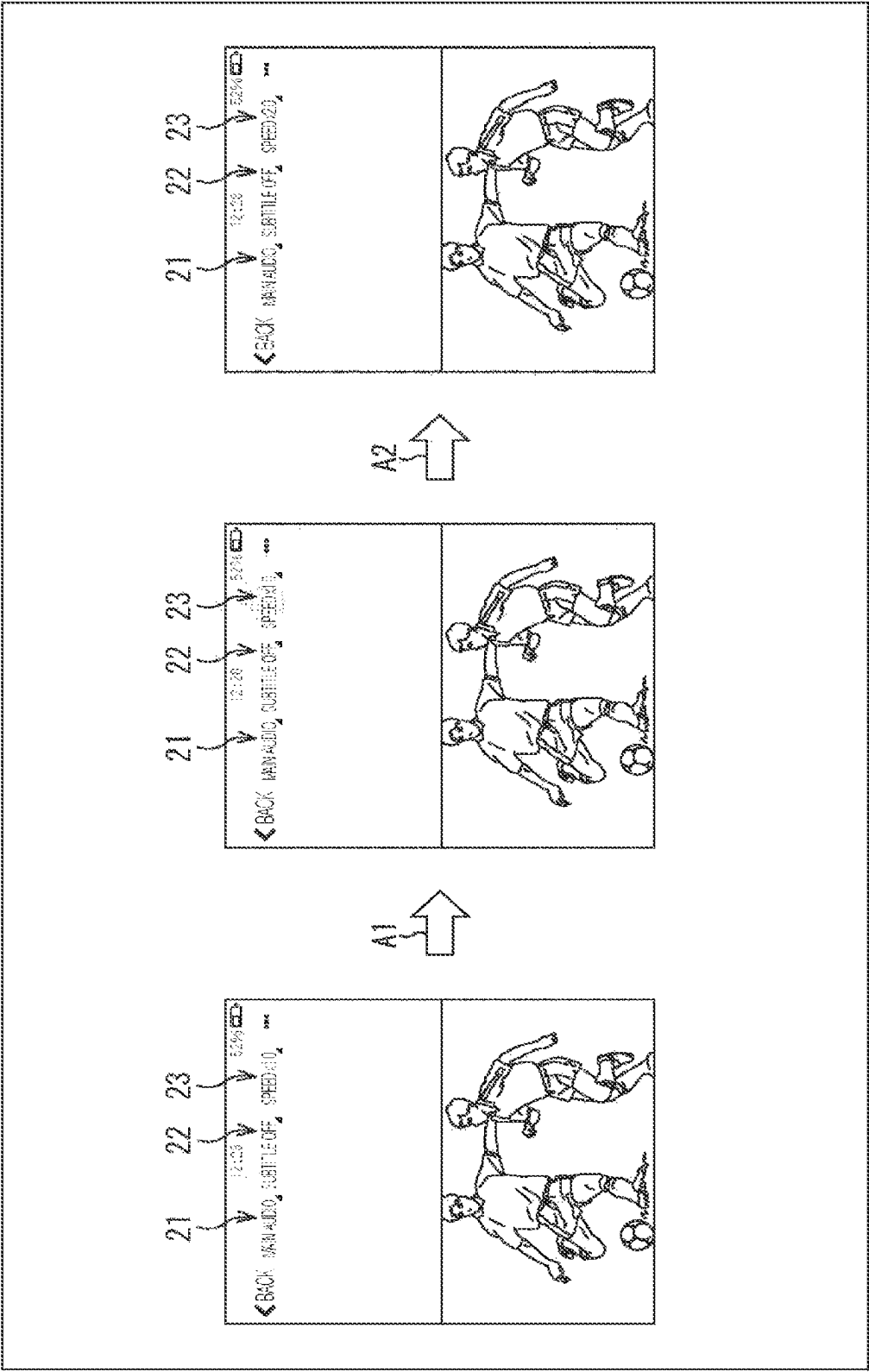


FIG. 6

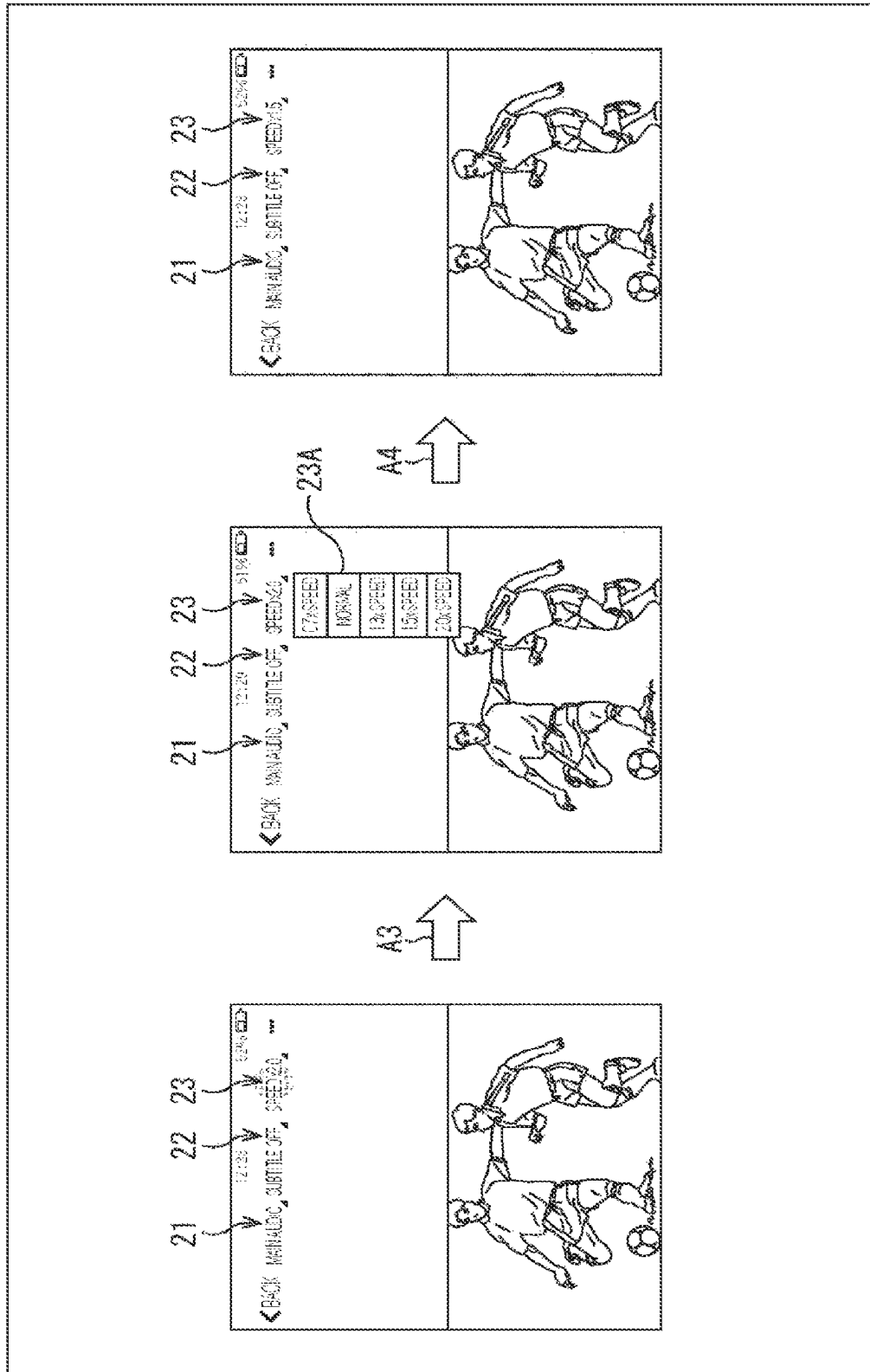


FIG. 7

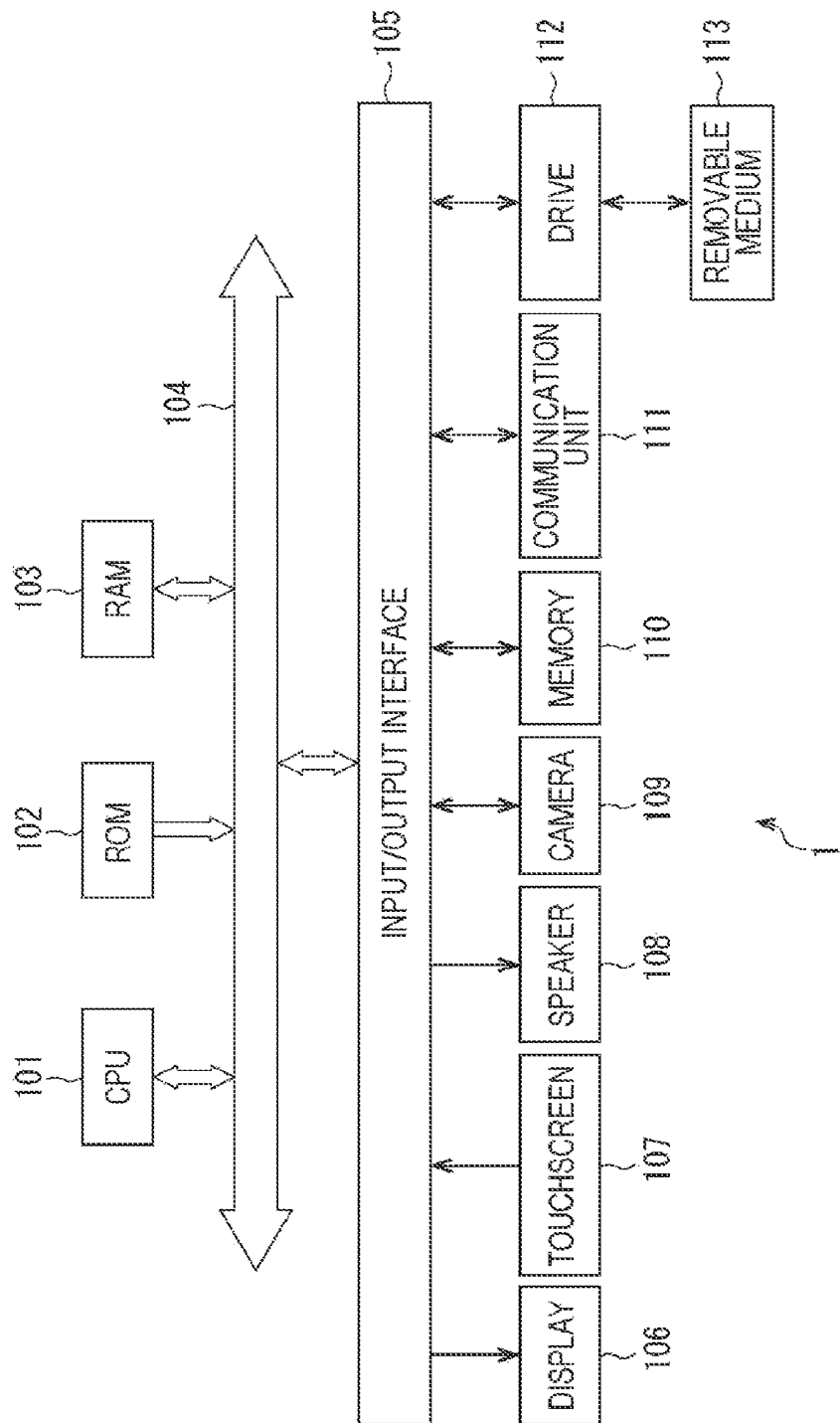


FIG. 8

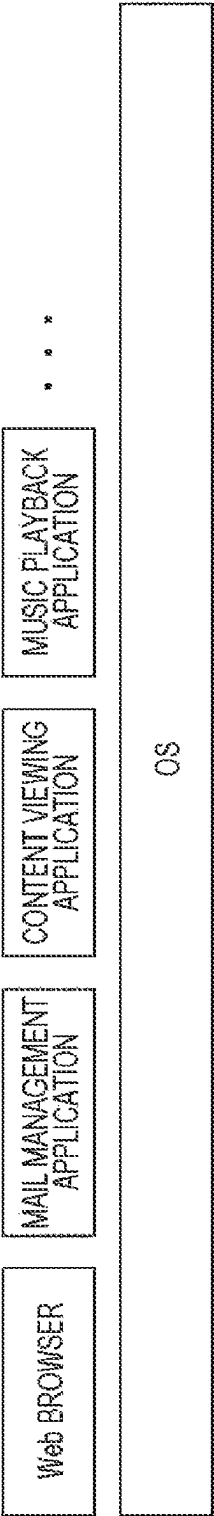


FIG. 9

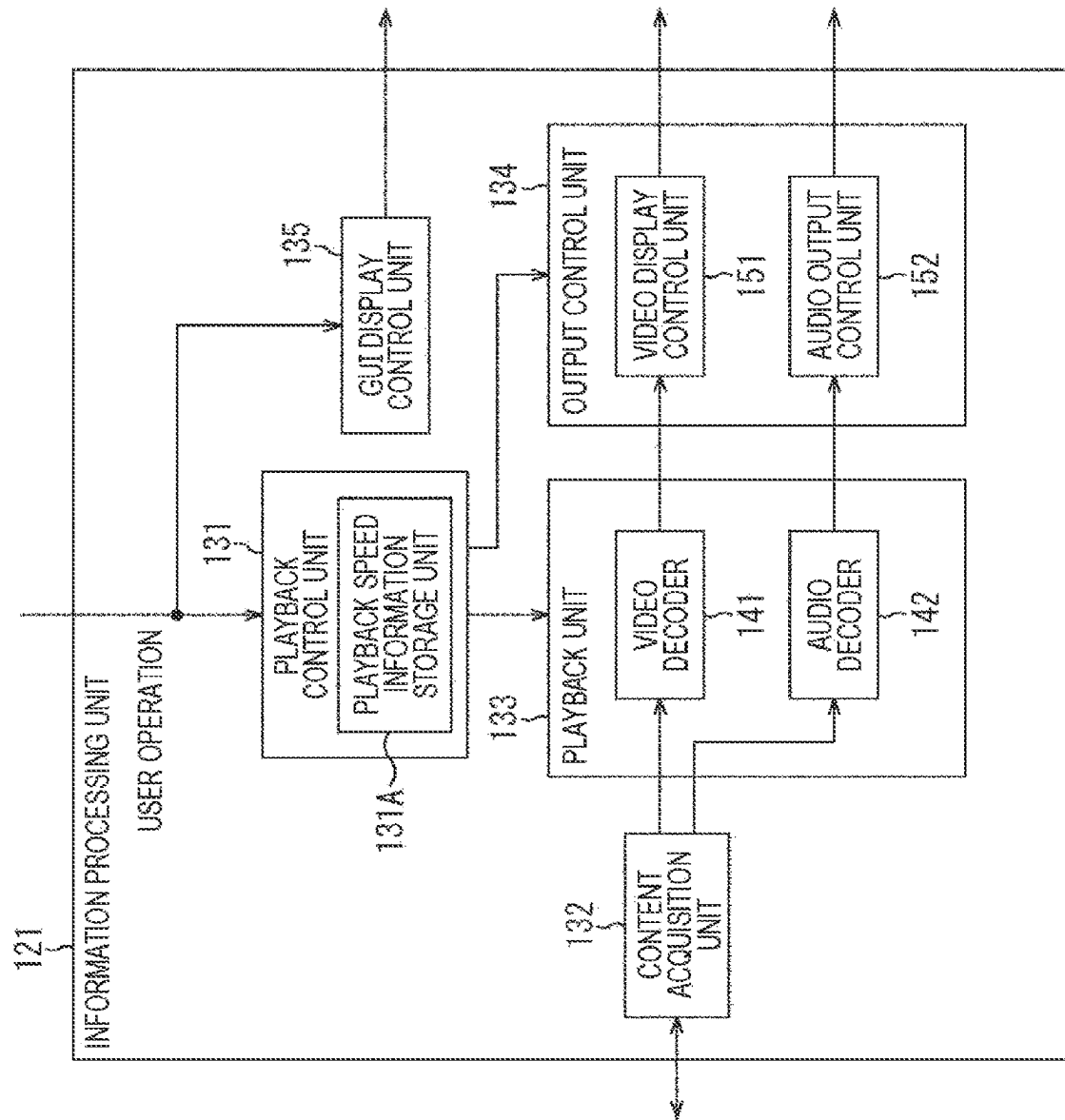


FIG. 10

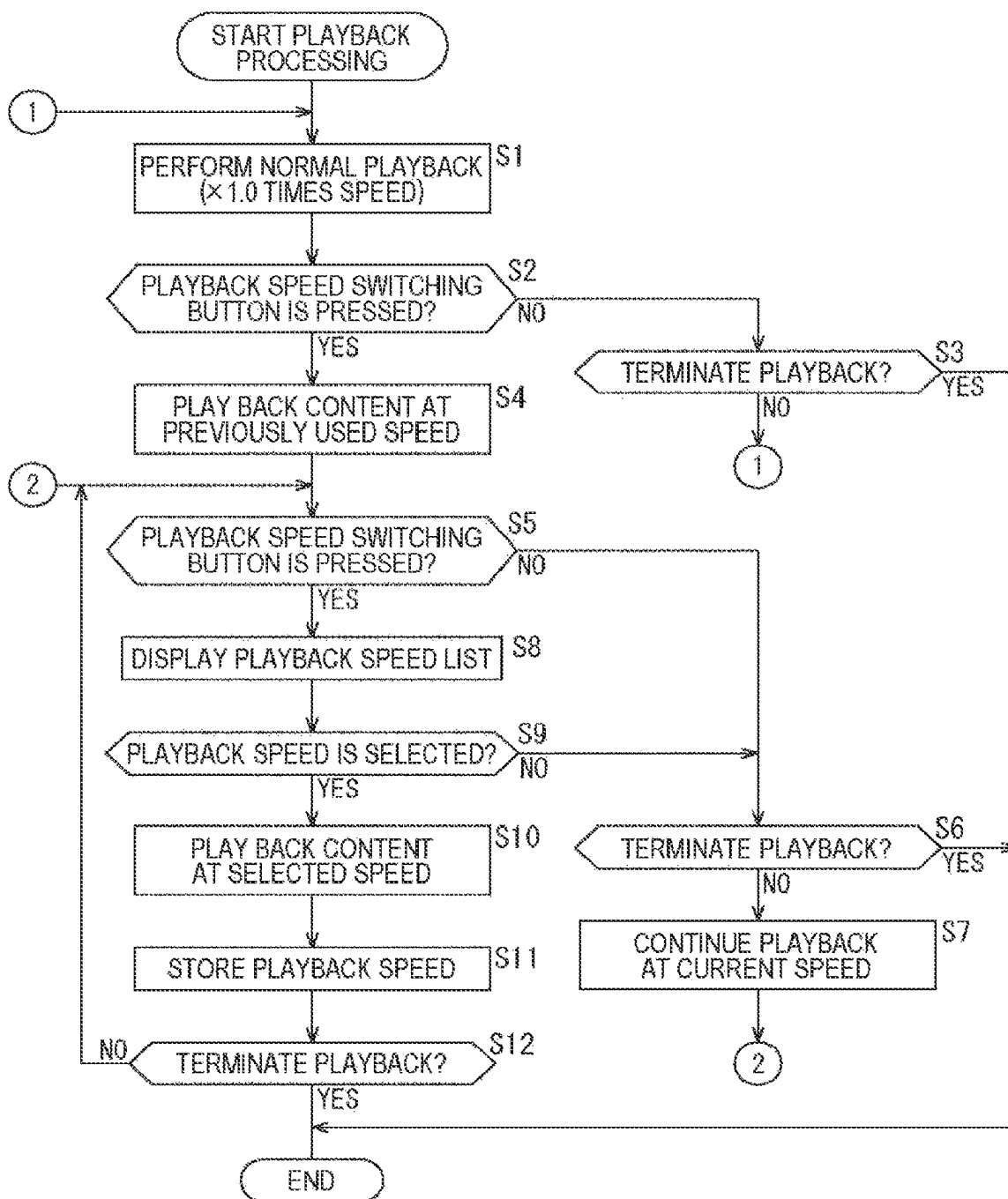


FIG. 11

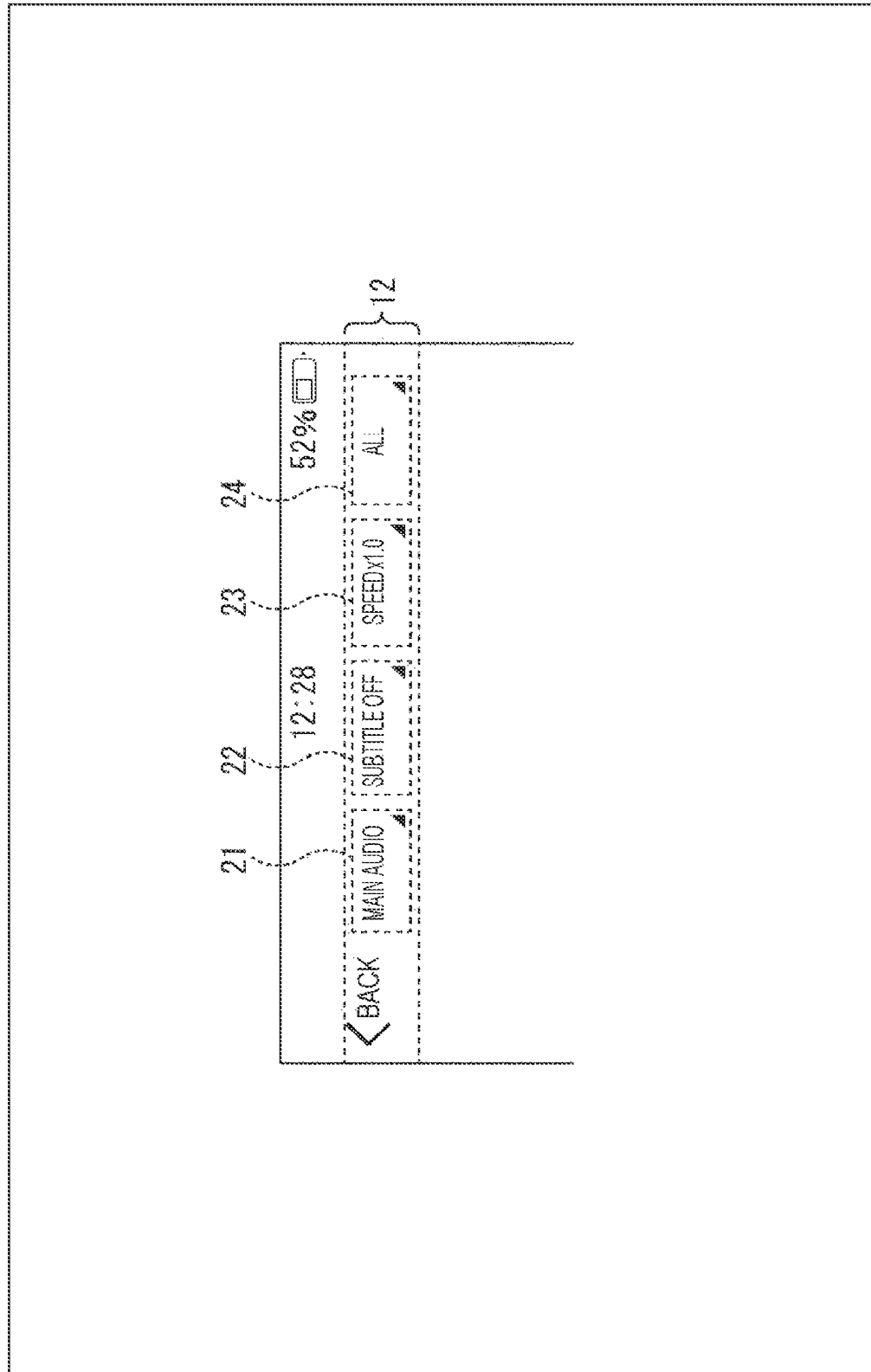
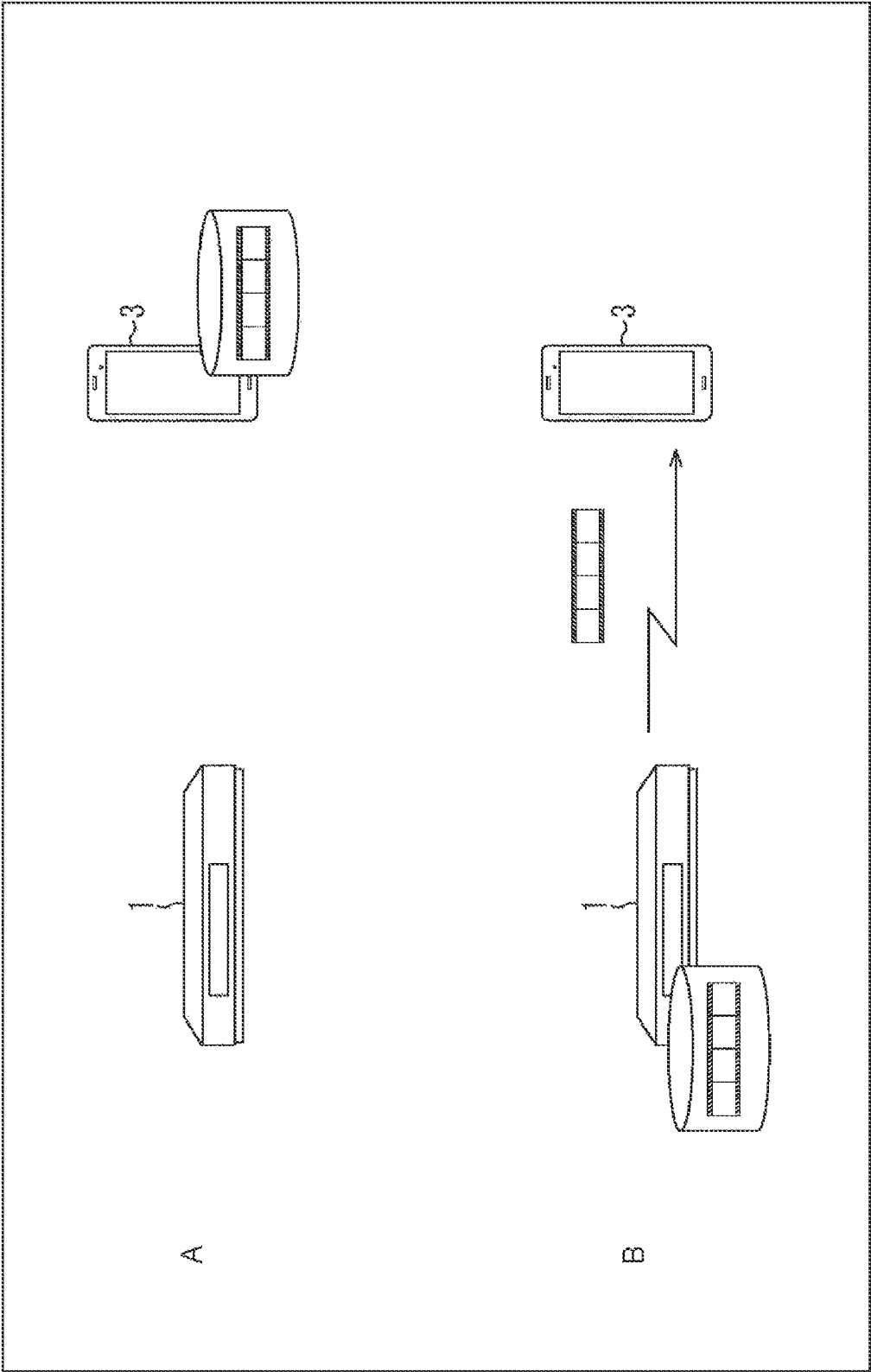


FIG. 12



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INFORMATION PROCESSING TERMINAL, INFORMATION PROCESSING METHOD, AND PROGRAM

This application is a Continuation of U.S. application Ser. No. 17/440,457, filed Sep. 17, 2021, which is a National Phase entry under U.S.C. § 371 of International Application No. PCT/JP2020/010744, filed Mar. 12, 2020, which claims priority benefit of Japanese Patent Application No. JP 2019-059130 filed in the Japan Patent Office on Mar. 26, 2019. Each of the above-referenced applications is hereby incorporated herein by reference in its entirety.

TECHNICAL FIELD

The present technology relates to an information processing terminal, an information processing method, and a program, and particularly relates to an information processing terminal, an information processing method, and a program, each of which makes it possible to easily switch a setting regarding playback of content.

BACKGROUND ART

There has been a widespread form in which a recorded program is transferred to a portable terminal such as a smartphone or a tablet terminal and is viewed by using the portable terminal. Many users view a previous day's news program during commute time or the like at a playback speed faster than 1x speed serving as a normal playback speed.

CITATION LIST

Patent Document

Patent Document 1: Japanese Patent Application Laid-Open No. 2018-117311

SUMMARY OF THE INVENTION

Problems to be Solved by the Invention

Many general UIs for use in switching a playback speed display a list of playback speed options and allow a user to select one from the options displayed as a list. Operation using such UIs forces the user to perform the following two-step operation, i.e., an operation for displaying a list of options and an operation for selecting one option.

The present technology has been made in view of such a situation, and makes it possible to easily switch a setting regarding playback of content.

Solutions to Problems

An information processing terminal according to an aspect of the present technology includes a control unit that starts playback of content by using a default value as a setting value of a predetermined setting regarding playback, and controls the playback of the content so that, in a case where a switching operation for switching the predetermined setting is performed, the content is played back by using a setting value selected during previous playback.

In an aspect of the present technology, playback of content is started by using a default value as a setting value of a predetermined setting regarding playback, and the playback of the content is controlled so that, in a case where

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a switching operation for switching the predetermined setting is performed, the content is played back by using a setting value selected during previous playback.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 illustrates a configuration example of a content playback system according to an embodiment of the present technology.

FIG. 2 illustrates an example of screen display of a content viewing application.

FIG. 3 is an enlarged view of a menu area.

FIG. 4 illustrates a display example of a playback speed list.

FIG. 5 illustrates an example of switching a playback speed by a one-step switching function.

FIG. 6 illustrates an example of switching a playback speed by a one-step switching function.

FIG. 7 is a block diagram illustrating a hardware configuration example of a portable terminal.

FIG. 8 illustrates an example of an application structure.

FIG. 9 is a block diagram illustrating a functional configuration example of a portable terminal.

FIG. 10 is a flowchart showing playback processing of a portable terminal.

FIG. 11 is another enlarged view of a menu area.

FIG. 12 illustrates examples of a location of content.

MODE FOR CARRYING OUT THE INVENTION

Hereinafter, modes for carrying out the present technology will be described. Description will be provided in the following order.

1. Configuration of content playback system
2. Switching playback speed in content viewing application
3. Configuration of portable terminal
4. Operation of portable terminal
5. Modification examples

<Configuration of Content Playback System>

FIG. 1 illustrates a configuration example of a content playback system according to an embodiment of the present technology.

The content playback system in FIG. 1 includes a recording device 1, a television receiver (TV) 2, and a portable terminal 3. The recording device 1, the TV 2, and the portable terminal 3 are devices owned by the same user, for example.

The recording device 1 receives content provided via various transmission paths, such as content provided via a broadcast wave, content provided via a CATV line, and content provided from a predetermined server via the Internet, and records the received content on a recording medium. The recording medium such as an HDD or an SSD is included in or externally attached to the recording device 1.

Content processed by the recording device 1 is content including video and audio, such as a television program provided by a broadcast station and moving image content distributed by a moving image distribution service. A plurality of pieces of content is recorded as recorded content on the recording medium of the recording device 1.

The recording device 1 has a function of playing back the recorded content and outputting video and audio of the content to the TV 2.

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Further, the recording device 1 has a function of transferring the recorded content to the portable terminal 3 via a network such as a wireless LAN or WAN.

The portable terminal 3 receives the content transferred from the recording device 1 and records the content on an internal recording medium such as a memory. The portable terminal 3 plays back the content recorded on the internal recording medium, displays the video of the content on a display, and outputs the audio from a speaker.

Thus, a content viewing application that is an application having a function of receiving and recording content transferred from the recording device 1 and playing back and outputting the content is installed in the portable terminal 3.

By starting the content viewing application and performing a predetermined operation, the user can transfer the content from the recording device 1 to the portable terminal 3 or can play back the content recorded on the internal recording medium of the portable terminal 3 and view the content on the portable terminal 3. The user can view the recorded content not only by using the

TV 2 but also by using the portable terminal 3 outside the user's house or the like.

The example of FIG. 1 illustrates a smartphone as the portable terminal 3, but other devices such as a tablet terminal and a PC can also be used as the portable terminal 3.

Switching Playback Speed in Content Viewing Application

FIG. 2 illustrates an example of screen display of the content viewing application.

As illustrated in FIG. 2, a video display area 11 serving as a display area of video of content is provided substantially at the center of a screen of the content viewing application. In the example of FIG. 2, video of a recorded soccer program is displayed in the video display area 11.

A narrow band-shaped menu area 12 is provided in an upper part of the screen of the content viewing application. In the menu area 12, buttons for use in settings regarding playback of the content are displayed side by side.

An operation panel area 13 is provided in a lower part of the screen of the content viewing application. In the operation panel area 13, buttons for use in operating the content, such as a playback start button, a pause button, and a skip button, and information regarding the content, such as a title, are displayed.

The display of the portable terminal 3 is provided with a touchscreen. The user can perform various operations by pressing the buttons displayed as a graphical user interface (GUI) in the menu area 12 and the operation panel area 13.

FIG. 3 is an enlarged view of the menu area 12.

As indicated by broken lines, an audio switching button 21, a subtitle switching button 22, and a playback speed switching button 23 are displayed in the menu area 12. The broken lines in FIG. 3 are illustrated for convenience of description and are not actually displayed.

The audio switching button 21 is a button for use in an audio setting. In a case where the audio switching button 21 is pressed, a list including "main audio" and "secondary audio" serving as setting values of the audio setting is displayed. The user can select, from the list, which of the main audio or the secondary audio is used as audio of the content. The main audio is selected in the example of FIG. 3.

The subtitle switching button 22 is a button for use in a subtitle setting. In a case where the subtitle switching button

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22 is pressed, a list including "ON" and "OFF" serving as setting values of the subtitle setting is displayed. The user can select whether subtitle display is on or off from the list. The subtitle display is selected to be off in the example of FIG. 3.

The playback speed switching button 23 is a button for use in a playback speed setting. In the example of FIG. 3, 1× speed serving as a normal playback speed is selected. "×1.0" displayed on the playback speed switching button 23 indicates that the 1× speed is selected.

FIG. 4 illustrates a display example of a playback speed list.

In a case where the playback speed switching button 23 is pressed at a predetermined timing such as during playback of the content, a playback speed list 23A serving as a list of setting values of the playback speed is displayed below the playback speed switching button 23 as illustrated in FIG. 4. In the example of FIG. 4, four types of setting values of "0.7× speed", "normal", "1.3× speed", "1.5× speed", and "2.0× speed" are displayed as options. The user can select a preferred playback speed from the playback speed list 23A as the playback speed of the content being played back.

Note that a colored circle on the playback speed switching button 23 in FIG. 4 indicates that the playback speed switching button 23 has been pressed. The same applies to the following drawings.

As described above, the content viewing application has a function of switching the playback speed of the content. By switching the playback speed of the content, the user can view fast playback/slow playback of the content.

Further, the content viewing application has a one-step switching function that is a function of easily switching the playback speed of the content to the same playback speed as a previously used playback speed.

The previously used playback speed is, for example, a playback speed selected by the user when the user has viewed content by using the content viewing application at the most recent time based on a current timing of viewing the content by using the content viewing application. The previously played back content and the currently played back content may be the same or different.

FIG. 5 illustrates an example of switching the playback speed by the one-step switching function.

Herein, it is assumed that the user has selected 2.0× speed (×2.0) as the playback speed at the previous playback of the content. Information indicating that the previously used playback speed is 2.0× speed is stored and is managed by the content viewing application.

In a case where an instruction is made to start playback of predetermined content, playback of the content selected by the user is started at a normal (default) playback speed as illustrated in a left screen of FIG. 5. "×1.0" indicating that the normal playback speed is selected is displayed on the playback speed switching button 23 in the left screen of FIG. 5.

In a case where the playback speed switching button 23 is pressed once as in a screen indicated by an arrow A1 during normal playback that is playback at a normal playback speed, the same playback speed as the previously used playback speed, i.e., 2× speed is selected as in a screen indicated by an arrow A2. The playback speed of the content is switched, and the playback of the content is continued at 2× speed serving as the previously used playback speed.

"×2.0" indicating that the same playback speed as the previously used playback speed is selected is displayed on the playback speed switching button 23 in the right screen of FIG. 5.

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Thus, the one-step switching function is a function of performing normal playback at the start of playback of the content and, in a case where the playback speed switching button **23** is pressed once, switching the playback speed of the content to the playback speed previously used by the user.

The user can switch the playback speed of the content to the previously used playback speed only by pressing the playback speed switching button **23** once. That is, the user does not need to perform a two-step operation of pressing the playback speed switching button **23** to display the playback speed list **23A** and selecting a predetermined playback speed in that state.

In the example of FIG. 5, the same illustration is displayed in the video display area **11** on each screen, but the video of the content actually changes with the lapse of time.

In a case where the playback speed switching button **23** is pressed once as in a left screen of FIG. 6 during the playback of the content at 2× speed serving as the previously used playback speed, the playback speed list **23A** is displayed as in a screen indicated by an arrow **A3**. The user can select a preferred playback speed from the playback speed list **23A**, such as a playback speed different from the previously used playback speed (currently selected playback speed).

Thus, the first pressing of the playback speed switching button **23** is an operation for switching the playback speed of the content to the same playback speed as the previously used playback speed, and the second pressing thereof is an operation for displaying the playback speed list **23A**.

For example, in a case where 1.5× speed is selected as the playback speed from the playback speed list **23A**, the playback speed of the content is switched as in a screen indicated by an arrow **A4**, and the playback of the content is continued at 1.5× speed serving as the playback speed. “×1.5” indicating that 1.5× speed is selected as the playback speed is displayed on the playback speed switching button **23** in the right screen of FIG. 6.

In a case where 1.5× speed is selected as the playback speed, information indicating that the previously used playback speed is 1.5× speed is stored and is appropriately used by the content viewing application when the next content is played back.

As described above, the user can easily switch the playback speed of the content to the previously used playback speed by using the one-step switching function. Processing of the content viewing application for switching the playback speed of the content as described above will be described later with reference to a flowchart.

Configuration of Portable Terminal

FIG. 7 is a block diagram illustrating a hardware configuration example of the portable terminal **3**.

A central processing unit (CPU) **101**, a read only memory (ROM) **102**, and a random access memory (RAM) **103** are connected to each other by a bus **104**.

The bus **104** is further connected to an input/output interface **105**. The input/output interface **105** is connected to a display **106**, a touchscreen **107**, a speaker **108**, a camera **109**, a memory **110**, a communication unit **111**, and a drive **112**.

The display **106** includes an LCD, an organic EL display, or the like. The display **106** displays such a screen of the content viewing application as described above.

The touchscreen **107** detects a user operation on a surface of the display **106** and outputs information indicating details of the user operation.

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The speaker **108** outputs various sounds such as audio and music of content.

The camera **109** captures an image in response to a user operation and outputs image data.

The memory **110** stores various data such as a program executed by the CPU **101** and content transferred from the recording device **1**.

The communication unit **111** is an interface for wireless communication. The communication unit **111** communicates with external devices such as the recording device **1** and a server connected via the Internet.

The drive **112** drives a removable medium **113** such as a memory card, writes data to the removable medium **113**, and reads data stored in the removable medium **113**.

FIG. 8 illustrates an example of an application structure.

As illustrated in FIG. 8, a plurality of applications operating on an operating system (OS) is installed in the portable terminal **3**. The example of FIG. 8 shows not only the content viewing application described above but also a web browser, a mail management application, and a music playback application.

The web browser displays a web page on the basis of information received from a server on the Internet. The mail management application is a mailer and manages sending and receiving of emails. The music playback application plays back music data stored in the memory **110** of the portable terminal **3** or music data received via the Internet, and outputs music from the speaker **108** of the portable terminal **3**.

Thus, the portable terminal **3** is a device in which various applications including the content viewing application can be installed to extend a function thereof.

FIG. 9 is a block diagram illustrating a functional configuration example of the portable terminal **3**.

In the portable terminal **3**, the CPU **101** in FIG. 7 executes the content viewing application to implement an information processing unit **121**. The portable terminal **3** is an information processing terminal including the information processing unit **121**.

As illustrated in FIG. 9, the information processing unit **121** includes a playback control unit **131**, a content acquisition unit **132**, a playback unit **133**, an output control unit **134**, and a GUI display control unit **135**.

The playback control unit **131** controls playback of content by the playback unit **133**. For example, the playback control unit **131** switches a playback speed of the content to the same playback speed as a previously used playback speed. A playback speed information storage unit **131A** included in the playback control unit **131** stores information indicating the playback speed previously used by the user.

Further, the playback control unit **131** controls output of the content by the output control unit **134**. Under the control of the playback control unit **131**, video of the content is displayed on the display **106**, and audio of the content is output from the speaker **108**.

The content acquisition unit **132** acquires data of content to be played back. For example, the content acquisition unit **132** acquires a piece of content selected by the user from pieces of content transferred from the recording device **1** and stored in the memory **110**.

The content acquisition unit **132** extracts video data from the data of the content and outputs the video data to a video decoder **141** of the playback unit **133**, and extracts audio data therefrom and outputs the audio data to an audio decoder **142**.

The playback unit **133** includes the video decoder **141** and the audio decoder **142**.

The video decoder **141** decodes the video data of the content, and outputs the decoded video data to a display control unit **151**.

The audio decoder **142** decodes the audio data of the content and outputs the decoded audio data to an audio output control unit **152**.

The output control unit **134** includes the display control unit **151** and the audio output control unit **152**.

The display control unit **151** causes the display **106** to display video of the content on the basis of the data supplied from the video decoder **141**.

The audio output control unit **152** causes the speaker **108** to output audio of the content on the basis of the data supplied from the audio decoder **142**.

As described with reference to FIGS. **5** and **6**, the

GUI display control unit **135** controls, in response to a user operation, display of the menu area **12** serving as a part of the screen of the content viewing application.

Operation of Portable Terminal

Herein, processing of the portable terminal **3** for playing back content will be described with reference to the flow-chart of FIG. **10**.

The processing in FIG. **10** is started when the content viewing application is activated and the content to be played back is selected by the user. Data of the content selected by the user is acquired by the content acquisition unit **132** and is supplied to the playback unit **133**.

In step **S1**, the playback control unit **131** controls the playback unit **133** to perform normal playback that is playback at a normal playback speed.

In step **S2**, the playback control unit **131** determines whether or not the playback speed switching button **23** has been pressed.

In a case where it is determined in step **S2** that the playback speed switching button **23** has not been pressed, the playback control unit **131** determines whether or not to terminate playback of the content in step **S3**.

In step **S3**, in a case where it is determined to terminate the playback of the content, the playback control unit **131** terminates the playback of the content, whereas, in a case where it is determined not to terminate the playback of the content, the processing returns to step **S1** and the normal playback is continued.

Meanwhile, in a case where it is determined in step **S2** that the playback speed switching button **23** has been pressed, in step **S4**, the playback control unit **131** switches the playback speed of the content and continues the playback of the content at the same playback speed as a previously used playback speed. The previously used playback speed is specified on the basis of information stored in the playback speed information storage unit **131A**.

The one-step switching function is achieved by the above processing, and the content is played back fast or slowly at the same playback speed as the playback speed previously used by the user.

In step **S5**, the playback control unit **131** determines whether or not the playback speed switching button **23** has been pressed again.

In a case where it is determined in step **S5** that the playback speed switching button **23** has not been pressed, the playback control unit **131** determines whether or not to terminate the playback of the content in step **S6**.

In step **S6**, in a case where it is determined to terminate the playback of the content, the playback control unit **131** terminates the playback of the content, whereas, in a case

where it is determined not to terminate the playback of the content, the processing proceeds to step **S7**, and the content is continuously played back at the current playback speed. Thereafter, the processing returns to step **S5**, and the above processing is repeated.

Meanwhile, in a case where it is determined in step **S5** that the playback speed switching button **23** has been pressed, the GUI display control unit **135** causes the playback speed list **23A** to be displayed in step **S8**.

In step **S9**, the playback control unit **131** determines whether or not a predetermined playback speed, such as a playback speed different from the currently selected playback speed, has been selected from the playback speed list **23A**. In a case where it is determined in step **S9** that the playback speed has not been selected, the processing proceeds to step **S6**, and the processing of step **S6** and subsequent steps is performed.

In a case where it is determined in step **S9** that the playback speed has been selected, in step **S10**, the playback control unit **131** switches the playback speed of the content and continues the playback of the content at the playback speed selected from the playback speed list **23A**.

In step **S11**, the playback control unit **131** causes information indicating the selected playback speed to be stored in the playback speed information storage unit **131A**. The playback speed indicated by the information stored in the playback speed information storage unit **131A** is used when the one-step switching function is operated during the next playback.

In step **S12**, the playback control unit **131** determines whether or not to terminate the playback of the content, and, in a case where it is determined not to terminate the playback of the content, the processing returns to step **S5** and the above processing is repeated.

Meanwhile, in a case where it is determined to terminate the playback of the content in step **S12**, the playback control unit **131** terminates the playback of the content.

With the above processing, the user can easily switch the playback speed of the content to the same playback speed as the previously used playback speed only by pressing the playback speed switching button **23** once.

By selecting a preferred playback speed once, the user can easily switch the playback speed of the content to the preferred playback speed when the content is played back the next or subsequent time.

The single playback speed switching button **23** can have both a function of a button for use in selecting fast playback and a function of a button for use in selecting slow playback.

Modification Examples

Examples Applied to Settings Other than Setting of Playback Speed

There has been described a case where the playback speed of the content is switched by the one-step switching function, but switching a setting by using the one-step switching function is applicable to various settings regarding the playback of the content, such as an audio setting and a subtitle setting.

The audio setting is switched by using the above-described audio switching button **21** (FIG. **3**). Further, the subtitle setting is switched by using the subtitle switching button **22**.

There will be described a case where a setting value of the audio setting is switched by the one-step switching function. Herein, it is assumed that the user has used secondary audio as a previous audio setting.

In this case, the content is played back by using, for example, main audio that is a normal audio setting at the start of playback of the content.

In a case where the audio switching button **21** is pressed once during the playback of the content using the main audio, the audio setting is switched to the secondary audio that is the previously used audio setting.

In a case where the audio switching button **21** is pressed again during the playback of the content using the secondary audio, an audio setting list is displayed, and either the main audio or the secondary audio can be selected. The audio selected herein is stored as information indicating the previously used audio setting.

The subtitle setting can be similarly switched by the one-step switching function.

The one-step switching function is applicable not only to the audio setting and the subtitle setting but also to various settings regarding the playback of the content such as an image quality setting and an audio quality setting.

A plurality of settings regarding the playback of the content may be collectively switched by the one-step switching function.

FIG. **11** is another enlarged view of the menu area **12**.

In the example of FIG. **11**, a button **24** is displayed to the right of the playback speed switching button **23**. The button **24** is a button operated when a plurality of settings regarding the playback of the content is collectively switched by the one-step switching function.

In a case where the button **24** is pressed once during the playback of the content using normal setting values, the setting values of the audio setting, the subtitle setting, and the playback speed setting are switched to the same setting values as setting values previously used by the user, respectively.

Thus, a plurality of settings may be collectively switched by the one-step switching function.

Control of One-Step Switching Function Depending on Location of Content

Enabling/disabling switching the setting of the playback speed by the one-step switching function may be controlled depending on a location (supply source) of the content.

FIG. **12** illustrates examples of the location of the content.

As illustrated in A of FIG. **12**, in a case where the content to be played back is recorded on the internal recording medium of the portable terminal **3**, switching the setting of the playback speed by the one-step switching function is enabled. As described above, the user can switch the playback speed of the content by using the one-step switching function.

Meanwhile, as illustrated in B of FIG. **12**, in a case where the content to be played back is recorded on the recording medium included in the recording device **1** and the content is viewed on the portable terminal **3** by streaming, switching the playback speed is disabled due to performance of the device or limitation on a communication speed. In this case, switching the setting of the playback speed by the one-step switching function is disabled.

Other

A case of playing back the content transferred from the recording device **1** and recorded in the portable terminal **3**

has been described, but switching a setting by the one-step switching function may be used when moving image content distributed by a moving image distribution service is played back. In this case, at least a part of the configuration of the information processing unit **121** in FIG. **9** is achieved in a server that provides the moving image distribution service.

A setting value to be switched by the one-step switching function may be different depending on a genre of the content being played back.

For example, in a case where an instruction is made to switch the setting of the playback speed by the one-step switching function, the playback speed is switched to 2× speed when the genre of the content being played back is “news”, and the playback speed is switched to 1× speed when the genre of the content being played back is “movie”.

Thus, the setting value to be switched by the one-step switching function may be individually selected by the user depending on an attribute of the content such as a genre.

The content to be played back is assumed to be content including video and audio, but the one-step switching function described above may be used when other pieces of content such as recorded audio are played back.

The operation for using the one-step switching function is assumed to be an operation of pressing the playback speed switching button **23** once, but the one-step switching function may be used by other operations such as a swipe operation and a gesture operation.

Regarding Program

The series of processing described above can be executed by hardware or can be executed by software. In a case where the series of processing is executed by software, a program constituting the software is installed in a computer incorporated in dedicated hardware, a general-purpose personal computer, or the like.

The program to be installed is provided by being recorded on the removable medium **113** of FIG. **7** including an optical disk (compact disc-read only memory (CD-ROM), digital versatile disc (DVD), or the like), a semiconductor memory, or the like. Further, the program may also be provided via a wired or wireless transmission medium such as a local area network, the Internet, or digital broadcasting. The program can be installed in the ROM **102** or the memory **110** in advance.

Note that the program executed by the computer may be a program in which the processing is performed in time series in the order described in the present specification, or may be a program in which the processing is performed in parallel or at a necessary timing such as when a call is made.

In the present specification, a system means a set of a plurality of components (apparatuses, modules (parts), and the like), and it does not matter whether or not all the components are included in the same housing. Therefore, a plurality of apparatuses included in separate housings and connected via a network and a single apparatus including a plurality of modules in a single housing are both systems.

The effects described in the present specification are merely illustrative and are not limited. Further, additional effects may be obtained.

The embodiments of the present technology are not limited to the above embodiments, and can be variously modified without departing from the gist of the present technology.

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For example, the present technology can have a configuration of cloud computing in which a single function is shared and jointly processed by a plurality of apparatuses via a network.

Further, each of the steps described in the above flowchart can be executed by a single apparatus, or can be executed by being shared by a plurality of apparatuses.

Furthermore, in a case where a single step includes a plurality of processes, the plurality of processes included in the single step can be executed by a single apparatus, or can be executed by being shared by a plurality of apparatuses.

Examples of Combination of Configurations

The present technology can also have the following configurations.

(1) An information processing terminal including a control unit that starts playback of content by using a default value as a setting value of a predetermined setting regarding playback, and controls the playback of the content so that, in a case where a switching operation for switching the predetermined setting is performed, the content is played back by using a setting value selected during previous playback.

(2) The information processing terminal according to (1), further including

a display control unit that causes a list for use in selecting a setting value to be displayed in a case where a same operation as the switching operation is performed during the playback of the content using the setting value selected during the previous playback.

(3) The information processing terminal according to (1) or (2), in which

the switching operation is an operation of pressing a button once.

(4) The information processing terminal according to (2), further including

a storage unit that stores information indicating a setting value selected from the list as information indicating the setting value selected during the previous playback.

(5) The information processing terminal according to any one of (1) to (4), in which

the control unit enables, depending on a supply source of the content, a function of playing back the content by using the setting value selected during the previous playback in a case where the switching operation is performed.

(6) The information processing terminal according to (5), in which

the control unit enables the function in a case where the supply source of the content is a recording medium included in the information processing terminal, and disables the function in a case where the supply source of the content is a recording medium of an external device connected via a network.

(7) The information processing terminal according to any one of (1) to (6), in which

in a case where the switching operation is performed, the control unit plays back the content by using a setting value selected during previous playback of content of a same genre as the content.

(8) The information processing terminal according to any one of (1) to (7), in which

the predetermined setting is a setting of a playback speed of the content including video and audio.

(9) An information processing method including causing an information processing terminal to

start playback of content by using a default value as a setting value of a predetermined setting regarding playback, and

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control the playback of the content so that, in a case where a switching operation for switching the predetermined setting is performed, the content is played back by using a setting value selected during previous playback.

(10) A program for causing a computer to execute processing of

starting playback of content by using a default value as a setting value of a predetermined setting regarding playback, and

controlling the playback of the content so that, in a case where a switching operation for switching the predetermined setting is performed, the content is played back by using a setting value selected during previous playback.

REFERENCE SIGNS LIST

- 1 Recording device
- 3 Portable terminal
- 121 Information processing unit
- 131 Playback control unit
- 131A Playback speed information storage unit
- 132 Content acquisition unit
- 133 Playback unit
- 134 Output control unit
- 135 GUI display control unit

The invention claimed is:

1. An information processing terminal comprising circuitry configured to

start playback of content by using a default value of a predetermined playback setting,

control the playback of the content so that, in a case where a switching operation for switching the predetermined playback setting is performed, the content is played back by using a playback setting selected during previous playback,

display a playback setting list in a case where another same switching operation is performed during the playback of the content using the playback setting selected during the previous playback,

wherein

the switching operation includes pressing a button once, and

the playback setting is a content genre setting.

2. The information processing terminal according to claim 1, further comprising a storage unit that stores information indicating a playback setting selected from the list as information indicating the playback setting selected during the previous playback.

3. The information processing terminal according to claim 1, wherein the circuitry is configured to enable, depending on a supply source of the content, playing back the content by using the playback setting selected during the previous playback in a case where the switching operation is performed.

4. The information processing terminal according to claim 3, wherein the circuitry is configured to enable playing back the content in a case where the supply source of the content is a recording medium included in the information processing terminal and disable playing back the content in a case where the supply source of the content is a recording medium of an external device connected via a network.

5. The information processing terminal according to claim 1, wherein in a case where the switching operation is performed, the circuitry is configured to play back the content by using a playback setting selected during previous playback of content of a same genre as the content.

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6. The information processing terminal according to claim 1, wherein the predetermined setting includes a playback speed of the content including video and audio.

7. An information processing method operating in an information processing terminal comprising

starting playback of content by using a default value of a predetermined playback setting,

controlling the playback of the content so that, in a case where a switching operation for switching the predetermined playback setting is performed, the content is played back by using a playback setting selected during previous playback,

displaying a playback setting list in a case where another same switching operation is performed during the playback of the content using the playback setting selected during the previous playback,

wherein

the switching operation includes pressing a button once, and

the playback setting is a content genre setting.

8. A non-transitory computer-readable medium having stored thereon a computer-readable program for causing a computer to execute processing of

starting playback of content by using a default value of a predetermined playback setting,

controlling the playback of the content so that, in a case where a switching operation for switching the predetermined playback setting is performed, the content is played back by using a playback setting selected during previous playback,

displaying a playback setting list in a case where another same switching operation is performed during the playback of the content using the playback setting selected during the previous playback,

wherein

the switching operation includes pressing a button once, and

the playback setting is a content genre setting.

9. The information processing method according to claim 7, further comprising storing information indicating a playback setting selected from the list as information indicating the playback setting selected during the previous playback.

10. The information processing method according to claim 7, further comprising depending on a supply source of the content, playing back the content by using the playback

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setting selected during the previous playback in a case where the switching operation is performed.

11. The information processing method according to claim 10, further comprising playing back the content in a case where the supply source of the content is a recording medium included in the information processing terminal, and disabling playing back the content in a case where the supply source of the content is a recording medium of an external device connected via a network.

12. The information processing method according to claim 7, further comprising in a case where the switching operation is performed, playing back the content by using a playback setting selected during previous playback of content of a same genre as the content.

13. The information processing method according to claim 7, wherein the predetermined setting includes a playback speed of the content including video and audio.

14. The information processing terminal according to claim 1, wherein the circuitry is configured such that the switching operation employs a one-step switching function.

15. The information processing terminal according to claim 1, wherein the circuitry is configured such that the switching operation employs a one-step switching function individually selected by a user depending on an attribute of the content.

16. The information processing terminal according to claim 14, wherein the circuitry is configured such that the one-step switching function is different depending on a genre of the content being played back.

17. The information processing terminal according to claim 1, wherein the circuitry is configured such that the switching operation employs a switching function that is different depending on a genre of the content being played back.

18. The information processing terminal according to claim 1, wherein the circuitry is configured such that a playback speed is reset to 1× speed when the playback speed is set to a speed that is not 1× when the switching operation involve switching from one genre to a different genre.

19. The information processing terminal according to claim 1, wherein the circuitry is configured such that the switching operation employs a swipe operation.

20. The information processing terminal according to claim 1, wherein the circuitry is configured such that the switching operation employs a gesture operation.

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